

Education

Carnegie Mellon University (CMU)

Ph.D. in Robotics

Pittsburgh, PA

Sept. 2021 - Now

Massachusetts Institute of Technology (MIT)

Bachelor of Science in Electrical Engineering and Computer Science, GPA:4.8/5.0

Cambridge, MA

Sept. 2014 - June 2019

Honors and Awards

CMU Presidential Fellow, funded by Paul and James Wang/Sercomm

May 2025

Best Poster Award, ICRA 2025 Dexterity in Multi-Fingered Hands Workshop

May 2025

42th International Physics Olympiad Gold Medal Winner

July 2011

Research

CMU RPL Lab

Ph.D. Student

Pittsburgh, PA

Sept. 2021 - Now

- My research is about superhuman tactile sensing, including actively estimating physical properties, estimating object pose during contact, and creating high-resolution 3D reconstructions. I am co-advised by Professor Michael Kaess and Professor Wenzhen Yuan.

ISEE, Inc.

Research Engineer

Cambridge, MA

Aug. 2019 - May 2021

- Invented an efficient planning algorithm with guaranteed bounds on the probability of safety violation in probabilistic dynamic environments [2]. Deployed real-time on a 12-tons autonomous truck. The work is supervised by Dr. Chris Baker and Professor Ying Nian Wu.

MIT Robust Robotics Group

Undergraduate Researcher

Cambridge, MA

Sept. 2018 - June 2019

- Designed novel Graph Neural Network (GNN) method and a novel graph representation of the observation history to help the autonomous robot navigate more efficiently without a map. The work is supervised by Professor Gregory J Stein and Professor Nicholas Roy.

NTHU Vision Science Lab

Research Assistant

HsinChu, Taiwan

Jan. 2017 - July 2017

- Invented a new robotics visual place recognition deep learning algorithm that can be used in topological map and navigation [1]. This work is supervised by Professor Min Sun.

Publications

- [10] **Hung-Jui Huang**, Amin Mirzaee, Wenzhen Yuan, Michael Kaess. *GelSLAM: A Real-Time, High-Resolution, and Robust 3D Tactile SLAM System*. [Ongoing Work]
- [9] Amin Mirzaee, **Hung-Jui Huang**, Wenzhen Yuan. *GelBelt: A Vision-based Tactile Sensor for Continuous Sensing of Large Surfaces*. IEEE Robotics and Automation Letters (RA-L). Jan. 2025
- [8] **Hung-Jui Huang**, Michael Kaess, Wenzhen Yuan. *NormalFlow: Fast, Robust, and Accurate Contact-based Object 6DoF Pose Tracking with Vision-based Tactile Sensors*. IEEE Robotics and Automation Letters (RA-L). Dec. 2024
- [7] Irving Fang, Kairui Shi, Xujin He, Siqi Tan, Yifan Wang, Hanwen Zhao, **Hung-Jui Huang**, Wenzhen Yuan, Chen Feng, Jing Zhang. *FusionSense: Bridging Common Sense, Vision, and Touch for Robust Sparse-View Reconstruction*. International Conference on Robotics and Automation (ICRA). Oct. 2024

- [6] Xinyuan Luo, Shengmiao Jin, **Hung-Jui Huang**, Wenzhen Yuan. *An Intelligent Robotic System for Perceptive Pancake Batter Stirring and Precise Pouring*. International Conference on Intelligent Robots and Systems (IROS). July 2024
- [5] **Hung-Jui Huang**, Jingyi Xiang, Wenzhen Yuan. *Kitchen Artist: Precise Control of Liquid Dispensing for Gourmet Plating*. International Conference on Robotics and Automation (ICRA). Nov. 2023
- [4] Xiaofeng Guo, **Hung-Jui Huang**, Wenzhen Yuan. *Estimating Properties of Solid Particles Inside Container Using Touch Sensing*. International Conference on Intelligent Robots and Systems (IROS). Oct. 2023.
- [3] **Hung-Jui Huang**, Xiaofeng Guo, Wenzhen Yuan. *Understanding Dynamic Tactile Sensing for Liquid Property Estimation*. Robotics: Science and Systems (RSS). June 2022.
- [2] **Hung-Jui Huang**, Kai-Chi Huang, Michal p, Yibiao Zhao, Ying Nian Wu, Chris L. Baker. *Planning on a (Risk) Budget: Safe Non-Conservative Planning in Probabilistic Dynamic Environments*. International Conference on Robotics and Automation (ICRA). May 2021.
- [1] Tsun-Hsuan Wang*, **Hung-Jui Huang***, Juan-Ting Lin, Chan-Wei Hu, Kuo-Hao Zeng, Min Sun. *Omnidirectional CNN for Visual Place Recognition and Navigation*. International Conference on Robotics and Automation (ICRA). May 2018.

*Equal Contribution

Working Experience

ISEE, Inc.

Research Engineer

Cambridge, MA

Aug. 2019 - May 2021

- Invented a patentable state estimation algorithm that fulfilled the high standard robustness requirement; it is now a part of our autonomous navigation system
- Implemented a fast robot mapping algorithm for robust obstacle avoidance task for the autonomous navigation system

NVIDIA, Corp.

Deep Learning Intern

Santa Clara, CA

June 2018 - Aug. 2018

- Coordinated a three person team to implement a C++ pedestrian tracking algorithm running on a Jetson TX2 with real life demo
- Redesigned an object detection algorithm under Project Isaac; portion of it was used for GTC Japan 2018

NTHU Vision Science Lab

Research Assistant

HsinChu, Taiwan

Jan. 2017 - July 2017

- Directed the robot place recognition and navigation research team under supervision of professor Min Sun, and demonstrated the new algorithm on real world robot to Taiwanese government officials

Teaching

CMU Computer Vision (16-720)

Teaching Assistant

Pittsburgh, PA

Jan. 2024 - Now

- Revised and led the Lucas-Kanade assignment.

MIT Robotics Science and Systems (6.141)

Teaching Assistant

Cambridge, MA

Feb. 2018 - May 2018

- Devised a final class project, which used deep learning to navigate a mobile robot indoors using RGB sensors as the only sensor input; verified its feasibility prior to assigning to students
- Restructure the lab materials and advised students on classical robotics algorithms; received a 6.5/7.0 rating from students after the class ended

Skills

- **Programming languages:** Python, C++, Java, Assembly31
- **Software Libraries:** Pytorch, TensorFlow, ROS, Isaac SDK, NumPy, OpenCV, MATLAB GPOP
- **Software:** Linux, Ansys Fluent, SolidWorks, Arduino, Onshape, OpenSCAD, Psoc5